Model Optimization and Tuning Report

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Project Name: Diabetic Retinopathy Classification Web Application

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1. Introduction

Model optimization is an important phase to improve the performance and generalization ability of the trained CNN model. This phase focuses on tuning hyperparameters, reducing overfitting, and improving classification accuracy.

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2. Hyperparameter Tuning

The following hyperparameters were tuned:

- Learning Rate Adjustment
- Batch Size Modification
- Number of Epochs
- Optimizer Selection

The Adam optimizer with an optimized learning rate produced stable and consistent results.

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3. Regularization Techniques

To prevent overfitting, the following techniques were applied:

- Dropout Layers
- Early Stopping
- Data Augmentation
- Validation Monitoring

Dropout helped reduce model dependency on specific neurons, improving generalization.

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4. Data Augmentation

Image augmentation techniques were used to improve dataset diversity:

- Image Rotation
- Horizontal Flip
- Zoom
- Rescaling

This helped improve robustness of the model against variations in retinal images.

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5. Performance Improvement

After optimization:

- Validation accuracy improved
- Overfitting reduced
- Model stability increased
- Loss curves became smoother

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6. Final Optimized Model

The optimized model was saved as:

dr_model.h5

This optimized model is integrated into the Flask web application for final deployment.

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7. Conclusion

Model optimization significantly enhanced the performance and reliability of the Diabetic Retinopathy classification system. The tuned model is efficient, stable, and suitable for real-time medical image analysis.